

PRD: Program Enrolment Automation Action

1. Overview

Today, the platform's automation engine allows admins to enrol matched users into individual **courses** based on trigger-condition rules. However, there is no equivalent capability for **programs** — logical groupings of courses with their own enrolment lifecycle, drip scheduling, and progression logic. This means admins who want to route users into a program by rule must do so manually or build fragile workarounds enrolling users into each constituent course separately.

This ticket introduces a new automation **action type** — "Enrol in Program" — that sits alongside the existing "Enrol in Course" action within the current automation builder. It calls the **existing live program enrolment service** (no new enrolment logic), supports two timing modes (Immediate / Specific date), enforces idempotent enrolment (silently skipping already-enrolled users), and logs every trigger attempt to the Activity tab.

The job this feature is hired to do: **let an admin set up a rule once and trust that the right people end up in the right program, at the right time, without manual intervention or double-enrolment risk.**

2. Jobs To Be Done

Main Job

When I define the audience for a program, I want matching users to be automatically enrolled based on a rule, so that I don't have to manually enrol each person and can trust the system handles timing, deduplication, and logging.

Job Map

Stage	What happens today (Courses)	Gap for Programs	Target state
Define	Admin creates an automation with trigger + filters	No "Program" option in the action type	Admin can select "Program" as an action target alongside "Course"
Locate	Admin searches for a course to attach to the action	Course search only	A unified search that returns both courses and programs, clearly differentiated
Prepare	Admin configures Enrolment timing (Immediate / After	N/A for programs	Admin configures a single Enrolment column: Immediate or

	delay), Due date, Frequency		Specific date (with date picker)
Confirm	Admin reviews the automation rule before saving	No program preview	Same review flow, now showing program name and enrolment timing
Execute	Engine fires on trigger match, enrolls user into course	No program enrolment path	Engine calls the existing program enrolment service
Monitor	Activity tab logs each trigger (user, automation, date)	No program trigger logs	Program triggers appear in the same Activity tab with identical schema
Resolve	Duplicate enrolment prevented (course-level)	No program-level idempotency	Silently skip already-enrolled users; log the skipped trigger attempt

Related Jobs

- **Manage program content drip:** Drip scheduling is already handled by the live program enrolment service. This action defers to that service — drip starts from the actual enrolment date (which may be a future "Specific date").
- **Audit automation activity:** Admins need a single place (Activity tab) to verify that automations are firing correctly for both courses and programs.
- **Maintain data hygiene:** Preventing double enrolment protects reporting accuracy and avoids confusing learner experiences.

Emotional & Social Dimensions

- **Confidence:** "I set this up once and it just works — I don't worry about people falling through the cracks."
- **Control:** "I can see every trigger in the Activity log, so I'm never in the dark about what the system did."
- **Competence:** "I look organised to stakeholders because programs are populated automatically and on schedule."
- **Trust:** "The system won't double-enrol someone — I don't have to babysit it."

3. Goals

1. **Parity with course enrolment automation:** Programs become a first-class action target in the automation builder, using the same trigger-condition-action model.
2. **Zero new enrolment logic:** The action delegates entirely to the existing live program enrolment service, including drip handling.

3. **Idempotent by default:** Re-firing an automation for an already-enrolled user silently skips enrolment but records the trigger attempt.
 4. **Minimal UI surface area:** A single Enrolment dropdown (Immediate / Specific date) with progressive disclosure of a date picker — no Due date or Frequency columns for programs.
 5. **Full auditability:** Every program automation trigger (successful enrolment or skipped duplicate) appears in the Activity tab.
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4. Job Stories

Configuration

- **When** I am building an automation rule to onboard new hires into a learning path, **I want to** select a program (not just individual courses) as the enrolment action, **so I can** route them into the full structured program with one rule instead of creating separate course enrolments.
- **When** I am choosing the enrolment timing for a program action, **I want to** pick "Immediate" or "Specific date" from a single dropdown, **so I can** decide whether users are enrolled right away or on a scheduled future date without navigating extra configuration.
- **When** I select "Specific date," **I want** a date picker to appear inline, **so I can** choose the exact enrolment date without switching screens or mentally tracking date formats.
- **When** I search for a program to add to my automation, **I want** the search results to clearly distinguish programs from courses, **so I can** confidently select the right item without confusion.

Execution & Safety

- **When** an automation fires and the matched user is already enrolled in the target program, **I want** the system to silently skip the enrolment, **so I can** trust that no user is double-enrolled regardless of how many times the rule triggers.
- **When** I set a "Specific date" enrolment for a user who doesn't exist yet, **I want** the enrolment record to be created with the future start date and activated once the user registers, **so I can** pre-configure automations ahead of cohort arrivals.

Monitoring

- **When** I review the Activity tab for an automation, **I want to** see every program trigger alongside course triggers (user, automation name, triggered date), **so I can** audit the automation's behaviour from a single log.

- **When** a trigger fires but enrolment is skipped due to duplication, **I want** the Activity log to still record the trigger attempt, **so I can** understand why enrolment counts differ from trigger counts.

Eligibility

- **When** I search for programs to attach to an automation action, **I want** only enabled (active) programs to appear in the results, **so I can** avoid accidentally routing users into draft or archived programs.

5. Requirements

Functional Requirements

ID	Requirement	Priority	Notes
FR-01	The automation Actions table supports a new action type: Program . Admins can add program actions alongside existing course actions within the same automation.	Must have	Extends the existing Actions table; does not replace course actions.
FR-02	A program search field allows admins to find and select a target program. Only programs with status Enabled are returned.	Must have	Mirror the existing course search UX. Clearly label results as "Program" to differentiate from courses (see BFM "Suggested field types" pattern).
FR-03	The program action row displays a single Enrolment column with a dropdown: Immediate (default) or Specific date .	Must have	No Due date or Frequency columns for programs — these are course-specific.
FR-04	When Specific date is selected, an inline date picker is revealed via progressive disclosure. Past dates are disabled/greyed out.	Must have	Follows NN/g and PatternFly date-input guidelines.
FR-05	When the automation fires, enrolment is executed by calling the existing live program enrolment service . No new enrolment logic is introduced.	Must have	Drip scheduling, content release, and progression are handled by the existing service.
FR-06	Enrolment is idempotent : if the matched user is already enrolled in the target program, the system silently skips enrolment. No error is surfaced to the admin.	Must have	Use an upsert/existence-check pattern before calling the enrolment service.

FR-07	Every trigger attempt — whether it results in enrolment or a silent skip — is logged in the Activity tab with: user, automation name, action (program enrolment), target program, triggered date, and result (enrolled / skipped-duplicate).	Must have	Append-only log; follows the same schema as course automation triggers.
FR-08	For Specific date enrolment, the system creates an enrolment record with a future start date. If the user does not yet exist, the enrolment activates upon user registration.	Must have	Aligns with the clarified behaviour: "The user may not exist yet; once registered they get enrolled."
FR-09	The program action can coexist with course actions in the same automation. An automation may contain multiple program actions and multiple course actions.	Should have	Enables complex onboarding flows (e.g., enrol in Program A + Course X).
FR-10	The program search field supports type-ahead with a minimum of 2 characters, consistent with the existing course search.	Should have	Performance: paginate or limit results to avoid overwhelming the dropdown.

6. Research & Best Practices

Synthesis

The automation builder follows the well-established **trigger → condition → action** pattern documented by [UI Patterns' Rule Builder reference](#) and exemplified in the [DelightChat automation rules case study](#). The existing course enrolment action already conforms to this structure; adding a "Program" action type is a natural extension rather than a new paradigm.

Progressive disclosure is the dominant UX principle for this feature. [NN/g's cognitive load article](#) identifies it as one of four key principles, and [PatternFly's date picker guidelines](#) specifically recommend showing date fields only when a scheduling option is selected. This directly informs the Enrolment dropdown design: default to "Immediate," and reveal the date picker only when "Specific date" is chosen.

For date input itself, [NN/g's date-input guidelines](#) advise that calendar pickers work well for dates within a year of the present — appropriate here since program enrolment dates are typically near-future. Past dates should be disabled and greyed out to prevent illogical scheduling.

Idempotency is critical. The [eLearning Industry article on LMS automation](#) warns explicitly that rerunning a workflow must not double-enrol a learner, recommending "already processed" checks and idempotent API calls. The [Dev.to article on idempotency patterns](#) provides concrete implementation guidance: upsert operations (INSERT or UPDATE if exists) are naturally idempotent, and a two-phase reservation pattern (atomically insert as IN_PROGRESS, then update to COMPLETED) eliminates race conditions during concurrent triggers — relevant if bulk imports match thousands of users simultaneously.

Audit logging requirements are informed by [AppMaster's audit logging guide](#), which recommends an append-only event log with actor, timestamp, action type, target entity, and JSON payload. For automation-triggered actions, the actor_type should be **system** with the automation name as the actor label. [GS Consulting's audit trail guidance](#) reinforces this with structured fields (event ID, correlation ID, workflow name, step, result) and append-only, restricted-write patterns.

The **competitive landscape** confirms this feature's necessity: [Kajabi already offers native automation rules tied to enrolment](#) with triggers based on quiz outcomes and assignment submissions, plus drip-release content. Teachable and Thinkific offer varying levels of program-level automation. Parity here is table stakes.

UX References

App	Flow / Screen	URL	Pattern	Relevance
Contractbook	Automation rule builder with action config	View	Trigger-condition-action rule builder with dropdown configuration	Direct parallel to the automation builder where the new program action will be added
Coda	Automation action configuration	View	Automation rule setup with action type selection and conditional fields	Shows how a SaaS tool structures action configuration within a rule builder
Calendly	Scheduling configuration with date/time options	View	Scheduling form with timing options and date picker	Reference for immediate vs scheduled timing selection in a configuration panel
ClickUp	Automation action dropdown and rule builder	View	Automation builder with action type selection	Pattern for adding new action types alongside existing ones in a rule builder
Kit	Automation rule builder with action config	View	Email/action automation configuration	Shows how an action row is configured with timing within an automation
Kajabi	Enrolment scheduling admin panel	View	LMS-style admin configuration for enrolment	Directly relevant: Kajabi is a course/program platform with enrolment scheduling
Teachable	Enrolment configuration in admin	View	Course/program admin setup with scheduling	Competitor LMS enrolment configuration pattern

Gusto	Scheduling config with immediate/date options	View	Admin panel with timing selection (now vs specific date)	Pattern for "immediate vs specific date" dropdown with conditional date picker reveal
Podia	Enrolment/membership admin config	View	Course platform admin enrolment settings	Competitor LMS enrolment flow reference

Built for Mars Lessons

Title	Type	URL	Lesson	Relevance
Avoiding duplicate payments	UX bite	View	Capital One detects and surfaces duplicate automatic payment conflicts, asking users to confirm or skip.	Directly relevant to idempotency: when the automation fires for an already-enrolled user, the system should silently skip but log the trigger — the right call for non-interactive automated actions.
Wait, is this a duplicate?	UX bite	View	Monzo warns users of identical transfers, showing time since last identical action.	Reinforces "records a trigger" requirement: even when skipping a duplicate enrolment, the attempt should be logged for auditability.
Hiding the advanced onboarding	UX bite	View	Stuff hides advanced options behind a collapsible section (progressive disclosure).	Supports the date picker pattern: default to "Immediate" and only reveal the date picker on "Specific date" selection, keeping the action row compact.
Suggested field types	UX bite	View	Notion suggests field types based on context, reducing configuration cognitive load.	When adding a new action row, the search/selector should clearly differentiate programs from courses with type labels or icons to prevent selection errors.
Progressively disclosing an upsell	UX bite	View	Impulse reveals options only after the user has context, avoiding information overload.	Confirms that the date picker should appear only after "Specific date" is chosen, not before — keeping the configuration row clean.

7. Plan of Action

Phase 1: Design & Specification

- ☐ Finalise UI mockups for the program action row in the Actions table, showing: program search field, single Enrolment dropdown (Immediate / Specific date), and conditionally revealed date picker.
- ☐ Design the program search component — type-ahead, "Program" type label/icon differentiation, filtered to enabled programs only.
- ☐ Specify the Activity log schema extension for program triggers (enrolled / skipped-duplicate result types).
- ☐ Document the API contract for the new action type: payload shape, validation rules (enabled program, valid future date), and response codes.
- ☐ Conduct design review with engineering to validate feasibility of reusing the existing program enrolment service without modification.

Phase 2: Backend Implementation

- ☐ Extend the automation action model to support a **program** action type alongside the existing **course** type.
- ☐ Implement the program enrolment action handler that calls the existing live program enrolment service.
- ☐ Implement idempotency check: before calling the enrolment service, query whether the user is already enrolled in the target program; if so, skip and log.
- ☐ Handle the "Specific date" case: create an enrolment record with a future start date; ensure the existing service activates it upon user registration if the user doesn't yet exist.
- ☐ Handle concurrent/bulk trigger scenarios: use an upsert or two-phase reservation pattern to prevent race conditions when many users match simultaneously.
- ☐ Extend the Activity log to record program automation triggers with the specified fields (user, automation, program, date, result).
- ☐ Add validation: reject action configuration if the selected program is not enabled (status check at save time and at trigger time).

Phase 3: Frontend Implementation

- ☐ Add "Program" as an action type option in the automation Actions table (alongside "Course").
- ☐ Build the program search component with type-ahead, enabled-only filtering, and clear "Program" labelling.
- ☐ Implement the Enrolment dropdown (Immediate / Specific date) with progressive disclosure of the date picker.
- ☐ Disable past dates in the date picker; default to Immediate.

- ☐ Display program automation triggers in the Activity tab using the same row format as course triggers.
- ☐ Ensure the force-trigger drawer (manual trigger for selected users) works with program actions.

Phase 4: Testing & QA

- ☐ Unit tests for the program enrolment action handler: successful enrolment, idempotent skip, future-date creation, non-existent user handling.
- ☐ Unit tests for validation: disabled/archived program rejection, past date rejection.
- ☐ Integration tests: end-to-end automation trigger → program enrolment → Activity log entry.
- ☐ Concurrency/load test: simulate bulk import matching thousands of users to verify no double-enrolments under race conditions.
- ☐ UI tests: action row rendering, progressive disclosure of date picker, program search filtering, Activity tab display.
- ☐ Regression tests: confirm existing course enrolment automations are unaffected.

Phase 5: Release & Rollout

- ☐ Feature flag the program action type for staged rollout.
- ☐ Update admin-facing documentation / help centre with the new action type.
- ☐ Monitor Activity logs post-launch for unexpected skip rates or errors.
- ☐ Collect admin feedback on the configuration UX after initial rollout cohort.

8. Risks & Mitigations

Risk	Impact	Mitigation
Race conditions during bulk triggers cause double-enrolment despite idempotency checks.	High — data integrity violation, corrupted reporting.	Use upsert or two-phase reservation pattern at the database level, as recommended by the idempotency patterns article . Include concurrency load tests in QA.
Program status changes after automation is saved (e.g., program is disabled/archived between config and trigger).	Medium — trigger fires against an invalid program, causing errors or silent failure.	Validate program status at trigger time (not just save time). Log a clear error in the Activity tab if the program is no longer enabled.

"Specific date" enrolment for non-existent users creates orphaned records if the user never registers.	Low — data clutter, minor storage cost.	Document expected behaviour. Consider a periodic cleanup job (out of scope for this ticket but flagged for future backlog).
Admin confusion between courses and programs in the search/selector.	Medium — wrong item selected, users enrolled in incorrect content.	Clearly differentiate programs from courses with type labels or icons in search results, per the BFM "Suggested field types" pattern.
Existing program enrolment service cannot handle high-throughput batch calls from automation triggers.	Medium — timeouts, partial enrolment of a matched cohort.	Profile the enrolment service under load before launch. Implement retry with backoff for transient failures; queue-based processing for large batches if needed.
Activity log volume increases significantly with program triggers added.	Low — performance degradation of Activity tab queries over time.	Append-only log with proper indexing. Paginate the Activity tab UI. Consider archival policy for old entries (future backlog).